

Advancing ASD Diagnostic Classification with Features of Continuous Wavelet Transform of fMRI and Machine Learning Algorithms

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Abstract— In this study, we aimed to develop a diagnostic classification model for autism spectrum disorder (ASD) using scalograms of functional magnetic resonance imaging (fMRI) data. Initially, the fMRI data from the ABIDE-I and ABIDE-II databases were preprocessed, and brain regions of interest (ROIs) were parcellated. Subsequently, scalograms were generated using Continuous Wavelet Transform from the average time series of each ROI. Various features were extracted from these scalogram images, including gray-level co-occurrence matrix, gray-level run-length matrix, fractal dimension texture analysis, Zernike's moments, Hu's moments, and first-order statistics. We employed CatBoost and Decision Tree machine learning classifiers and performed extensive five-fold cross-validation with hyperparameter tuning to evaluate the model's performance. The results indicated an overall accuracy of 83.04%, sensitivity of 66.66%, specificity of 85.57%, and an F1-score of 75.85% for the Decision Tree classifier, particularly a region in the Retro-Splenial Temporal network. In addition, regions from networks such as fronto-parietal task control network, Ventral Attention, and Visual, showcased promising classification performance for ASD. These findings highlight the efficacy of the proposed model in providing accurate ASD classification, which holds significant implications for early diagnosis and intervention strategies.

Keywords— autism spectrum disorder, fMRI, scalograms, feature extraction, machine learning

I. INTRODUCTION

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental condition that emerges during childhood and is characterized by three primary features: difficulties in forming social relationships, challenges in communication, and repetitive behavioral patterns. These deficits are believed to be linked to impaired executive functions, including working memory, inhibition, mental flexibility, and planning. ASD affects approximately 1 in 100 children worldwide, and due to its diverse nature, personalized therapies are necessary since there is no cure [1]. Detecting and treating ASD early are crucial steps in reducing symptoms and improving the quality of life for affected individuals. However, currently, there are no medical tests available for the detection of autism. Instead, ASD symptoms are typically identified through observation, with parents, teachers, and medical professionals such as pediatricians, child psychologists, neurologists, and therapists involved in the evaluation process. In adults,

identifying ASD symptoms can be more challenging than in older children and adolescents, as some symptoms may overlap with other mental health disorders. Despite behavioral changes in children as early as 6 months of age, the diagnostic process remains complicated due to the wide range of behavioral patterns and subjective symptoms associated with ASD, leading to delays in diagnosis that can extend from months to years. Consequently, there is a need to explore autism-specific brain imaging biomarkers to facilitate early detection of ASD [2].

Magnetic resonance examination is a valuable tool for investigating brain structure changes in children with ASD. Functional brain imaging, particularly functional magnetic resonance imaging (fMRI), has been extensively utilized to study overall brain functioning. These advanced neuroimaging techniques can detect subtle alterations in brain metabolism and neural network connectivity in specific regions over time. These changes can be induced by cognitive tasks or can be observed during the brain's spontaneous activity at rest. Blood-oxygen-dependent level (BOLD) fMRI, which measures changes in deoxyhemoglobin concentration, allows visualization of these neural metabolism changes related to cognitive tasks or spontaneous brain activity [3]. In this study, we utilized resting state-fMRI data to identify specific brain biomarkers associated with ASD.

Functional Connectivity (FC) is a widely used technique to compute the statistical relationship between BOLD signals in different brain regions. However, it's important to acknowledge that the BOLD signal is noisy and only partially reflects neural activity. It arises from various physiological processes like neuronal, metabolic, cardiac, and vigilance functions, and can be affected by machine and participant-specific characteristics. In FC analyses, these confounding effects can be particularly troublesome because they can introduce biased FC estimates due to high correlations between noise sources in different brain regions [3]. Resting-state activity in fMRI also shows nonstationary BOLD signal fluctuations, which create challenges when using traditional methods like Discrete Fourier transform (DFT) or short-time Fourier transform (STFT) for time-frequency (TF) analysis. DFT requires stationarity of the brain signal, which is not always the case, and STFT can be inconvenient due to the number of samples and sample frequency required [4]. continuous Wavelet Transform (CWT) allows for optimal

temporal resolution, representing the signal's TF structure as amplitudes and energy across various wavelet coefficients, enabling the tracking of frequency sub-bands over time [5]. In this study, we used a wavelet transform, specifically CWT, to represent BOLD fMRI time series signals in the TF domain.

The extraction of texture and moment-based features from scalograms is essential in identifying specific patterns of fMRI BOLD signal associated with ASD. Techniques like Gray Level Co-occurrence Matrix (GLCM) and Gray-Level Run-Length Matrix (GLRLM) are commonly used to quantify texture information in images. GLCM captures pixel spatial relationships, while GLRLM characterizes the lengths and orientations of pixel runs with the same gray level. Furthermore, methods such as Fractal Dimension Texture Analysis (FDTA), Zernike's moments (ZM), Hu's moments (HM), and first-order statistics (FOS) have been explored for feature extraction from 2D representations of signals. FDTA assesses complexity and irregularity in patterns, making it suitable for extracting texture features. ZM and HM provide global and local shape information in images, respectively. FOS, which includes mean, variance, skewness, and kurtosis, has been used as features from physiological signals for biomedical applications. In our study, we investigated the potential of using GLCM, GLRLM, FDTA, ZM, HM, and FOS based features from scalograms for ASD diagnosis.

Machine learning (ML) algorithms have been widely used to classify ASD based on handcrafted features extracted from fMRI signals. While many ML algorithms have been explored for this purpose, the decision tree algorithm has received less attention. Decision tree classifiers work hierarchically, using features to classify data at each level. Another algorithm, CatBoost, is a form of gradient boosting on decision trees that effectively handles categorical features, addressing gradient bias and prediction shift issues. On the other hand, deep learning models like convolutional neural networks (CNN) and recurrent neural networks have been used for diagnostic classification of ASD. However, these models face challenges with high-resolution images, requiring a large amount of data and numerous adjustable parameters during training. Additionally, a study has demonstrated that ML algorithms perform better when given texture patterns extracted from scalograms of signals rather than using scalograms directly supplied to CNN [6]. In this study, we utilized ML techniques, specifically decision trees and CatBoost, to construct a diagnostic classification model for ASD using handcrafted features from fMRI signals.

In our research, we have developed an automated diagnosis pipeline for ASD using fMRI and advanced ML techniques. The pipeline involves generating scalograms of the fMRI BOLD signal using CWT, extracting relevant features from these scalograms, and constructing classification models using DT and CatBoost algorithms. Our evaluation process involved testing various features and hyperparameters for different algorithms to gauge their performance. The promising results of our study have the potential to significantly enhance the accuracy and efficiency of ASD diagnosis, making this method highly applicable in real-world scenarios.

II. METHODOLOGY

A. Dataset

In this study, we obtained fMRI data from the publicly accessible Autism Brain Imaging Data Exchange (ABIDE-I and ABIDE-II) database. These databases contain numerous resting-state fMRIs, sMRIs, and DTIs collected from various locations worldwide, with approval from local institutional review boards [7]. We considered a participant pool of 36 individuals diagnosed with ASD and 35 typically developing (TD) individuals from Oregon Health and Science University (OHSU) site. Subject diagnostic status and demographic information were available on ABIDE's website and are presented in Table I.

TABLE I. DATASET DESCRIPTION

Parameters	ASD	TD
Number of subjects	36	35
Age in years (range)	13.57 $\pm$ 2.65 (7-15)	13.57 $\pm$ 2.65 (8-14)
Gender (M/F)	30/6	22/13
Handedness (R/L/M)	35/0/1	35/0/0
FIQ/VIQ/PIQ	106.19 $\pm$ 16.91	118.45 $\pm$ 12.17

M-Males, F-Females, R-Right, L-Left, M- Mixed handed, FIQ: Full Scale Intelligence Quotient, VIQ: Verbal Intelligent Quotient, PIQ: Performance Intelligence Quotient

B. Process Pipeline

Fig. 1 illustrates the proposed pipeline for ASD classification, consisting of six key processing stages: (1) Selection of fMRI data based on inclusion criteria, (2) Pre-processing, (3) Parcellation and extraction of BOLD time series, (4) Generation of scalograms, (5) Feature extraction, and (6) Model training, classification, and validation.

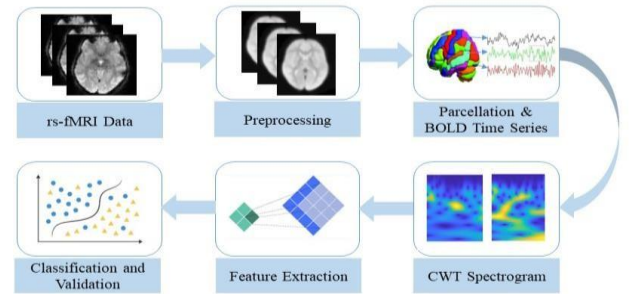


Fig. 1. Process pipeline of the proposed framework

C. Preprocessing

After applying standard preprocessing to the data, a whole-brain mask was generated to include only those brain voxels with consistent BOLD signals in at least 95% of the participants. The analysis focused on 236 regions of interest (ROIs) selected from three different atlases: 215 cortical ROIs from Gordon's atlas, 14 subcortical ROIs from the Harvard Oxford atlas, and 7 cerebellar ROIs from Diedrichsen atlases. We calculated average time series data in each of the 236 ROIs for each subject.

D. Time-frequency scalogram

Spectral analysis was performed using CWT. scalogram images were generated for each subject and each ROI's average time series data, resulting in a total of 16,756 images (236 ROIs $\times$ 71 Subjects). The CWT scalograms represent the frequency spectrum at each time point, considering the

frequency content within a specific neighborhood defined by the wavelet's width [8]. CWT $X_{cwt}(a, b)$ of a signal $x(t)$ is given by

$$X_{cwt}(a, b) = \int_{-\infty}^{\infty} x(t) \psi_{a,b}^*(t) dt \quad (1)$$

where $\psi_{a,b}^*(t)$ is a complex conjugate of scaled and shifted version of mother wavelet $\psi(t)$, a is scaling parameter and b is shifting parameter.

E. Feature Extraction

We extracted GLCM, GLRLM, ZM, HM, FDTA, and FOS features from each scalogram image for all participants. GLCM calculates joint probabilities of matching gray-level values between two pixels at different distances, providing information about the complexity, homogeneity, correlation, and local contrast of the image. GLRLM assesses spatial relationships between groups of pixels with similar gray-level values, enabling the evaluation of texture coarseness in a specified direction [9]. ZM represents a specific shape feature or pattern in the image. The zeroth-order ZM ($n=m=0$) represents the overall image intensity or the area of the binary region. Higher-order ZM captures more complex shape information, including orientation, symmetry, and curvature. The 2D ZM, Z_{nm} , of order n with repetition m , is defined in polar coordinates (r, θ) inside the unit circle as

$$Z_{nm} = \frac{n+1}{\pi} \int_0^{2\pi} \int_0^1 R_{nm}(r) e^{-jm\theta} f(r, \theta) r dr d\theta \quad (2)$$

where $R_{nm}(r)$ is the n th order of Zernike radial polynomial [10].

HM is computed from the normalized central moments of binary images or ROI to derive seven-moment invariants. These invariants serve as numerical descriptors for shape characterization, offering valuable information about overall geometry, compactness, and pixel intensity distribution. FDTA, on the other hand, involves applying fractal geometry principles to quantify the complexity and self-similarity of textures in images. This method helps measure the irregularity and detailed structure of textures. The FOS features include the mean, median, and standard deviation of pixel values, as well as descriptors characterizing the histogram of image pixel intensity values such as variance, kurtosis, skewness, and entropy [11][12][13][14]. The list of texture and shape-based features analyzed in this study is given in Table II.

TABLE II. LIST OF FEATURES

Texture/Shape analysis	Features
GLCM (38)	Contrast, Dissimilarity, Homogeneity, Energy (EN), Correlation, Angular Second Moment, Entropy, Mean, Variance, and Sum of Squares. The mean values and the range of values of the following features: 1) Angular Second Moment, 2) Contrast, 3) Correlation, 4) Sum of Squares Variance (SSV), 5) Inverse Difference Moment, 6) Sum Average, 7) Sum Variance, 8) Sum Entropy, 9) Entropy, 10) Difference Variance, 11) Difference Entropy, 12) Maximal

	Correlation Coefficient, 13,14) Information Measures of Correlation
GLRLM (5)	Short Run Emphasis (SRE), Long Run Emphasis (LRE), Grey Level Uniformity (GLU), Run Length Uniformity (RLU), and Run Percentage (RPC)
ZM (24)	Moments 1 to 24
HM (7)	Moments 1 to 7
FDTA (4)	Hurst Coefficients 1 to 4
FOS (7)	Mean, Variance, Skewness, Kurtosis, Energy, Entropy, and Coefficient of Variation

F. Classification

For the ML classification, two classifiers, CatBoost and DT, were employed. Optionally, feature scaling was performed using the StandardScaler method to normalize the input data. To assess the classifiers performance and prevent overfitting, k-fold cross-validation with hyperparameter tuning was carried out. The dataset was split into 5 folds, and for each classifier (CatBoost and DT), hyperparameters were tuned using GridSearchCV. The best hyperparameters were chosen based on accuracy, and the final classifiers were trained using the entire scaled dataset.

G. Evaluation metrics

The classifiers performance was assessed using several metrics, which included accuracy, sensitivity, specificity and F1 score. Accuracy represents the overall correctness of the model's predictions. Sensitivity measures the model's ability to correctly identify positive cases, while specificity indicates its accuracy in identifying negative cases. Precision reflects the model's effectiveness in identifying true positives and avoiding false positives. The F1 score, a combined metric of precision and recall, offers a comprehensive evaluation of the model's performance in handling both false positives and false negatives. This metric is particularly valuable when dealing with imbalanced data, where both types of errors are significant [15].

TABLE III. LIST OF METRICS AND THEIR FORMULA

Parameter	Formula
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$
Sensitivity	$TP/(TP+FN)$
Specificity	$TN/(TN+FP)$
F1 score	$2 * (Precision * Sensitivity) / (Precision + Sensitivity)$

TP-True Positive, TN-True Negative, FP-False Positive, FN-False Negative

III. RESULTS

Fig. 2 illustrates the average time series fMRI data for brain regions in individuals with ASD and TD subjects. Comparatively, the average voxel intensity in the ASD group is higher than in the TD group, although this difference may not be uniform across all subjects and regions being studied. Additionally, individuals with ASD demonstrate irregular or less synchronized fluctuations in neural activity, resulting in unique patterns in the TF representation when compared to TD subjects.

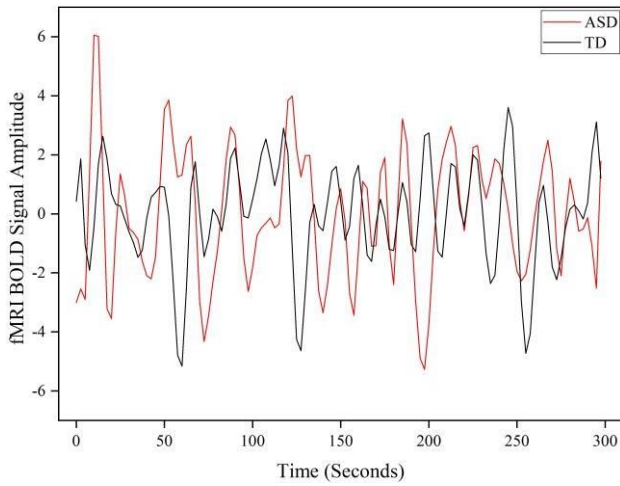


Fig. 2 Representation of average time series BOLD fMRI signal of a brain region of ASD and TD.

Fig. 3 displays the scalogram images obtained using CWT for the fMRI BOLD signal of individuals with ASD and TD subjects. Notably, fluctuations can be observed at the start and end of the fMRI BOLD signals of ASD and TD in the respective scalograms. On the other hand, varying patterns of fluctuations throughout the entire fMRI BOLD signals can also be observed in the scalograms of TD subjects.

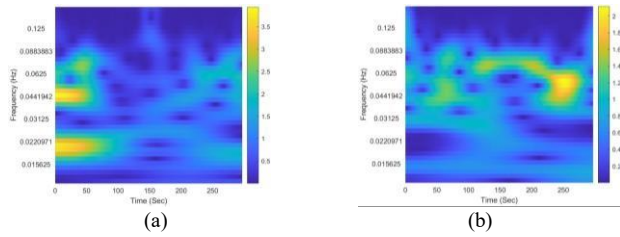
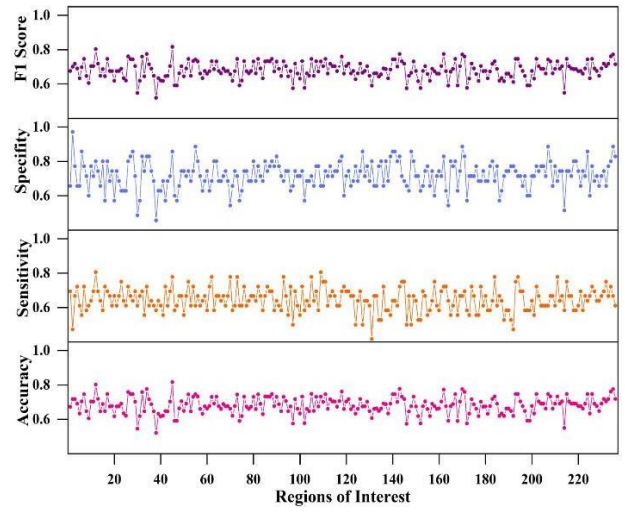
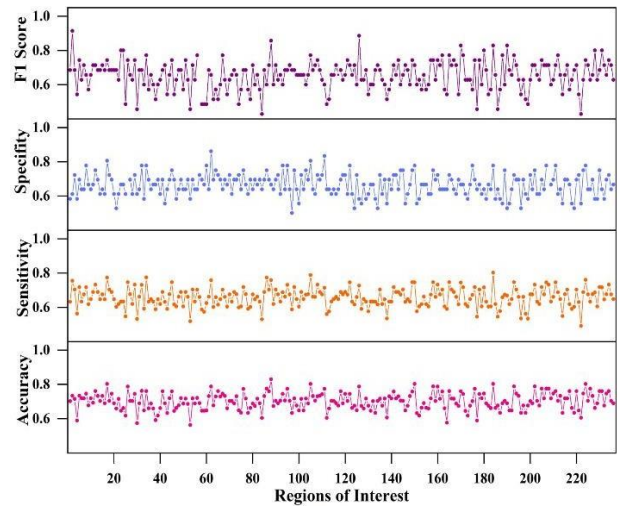


Fig. 3 Scalogram representation of fMRI BOLD signal of the subject using CWT (a) ASD and (b) TD

In Fig. 4 (a) and (b), we present the results of our classification experiment using CatBoost and DT models. We extracted texture and shape features from the scalogram of 236 brain regions and used them as input. Notably, the DT model consistently outperformed CatBoost across various brain regions. The highest classification accuracy achieved was 83.04% with DT, and its corresponding sensitivity, specificity, and f1-score were 66.66%, 85.57%, and 75.85%, respectively. On the other hand, CatBoost yielded the highest accuracy of 81.61%, along with sensitivity, specificity, and F1 score of 77.77%, 85.71%, and 81.66%, respectively. It is important to mention that the classification performance varied significantly across different brain regions and their corresponding networks. Some regions showed better classification performance, indicating their higher importance for analysis compared to considering all ROIs. This approach not only saves computational time but also enables clinicians to focus on specific brain areas that have a more pronounced impact on ASD.



(a)



(b)

Fig. 4. Average 5-fold classification performance using (a) CatBoost and (b) DT

Table 4 presents the top five brain regions that achieved the highest classification accuracies using catBoost and DT, along with their corresponding coordinates and networks. The brain networks that significantly contributed to the classification accuracy were not consistent between the two classifiers. However, a few networks showed consistency within each classifier. Specifically, regions from the COTC and SMH networks appeared twice among the top five models of CatBoost, while the VA network contributed twice to the top five models of DT. In conclusion, the brain networks COTC, SMH, RST, VA, and Visual displayed better results for the classification model, demonstrating their importance in distinguishing between individuals with ASD and typically developing subjects.

TABLE IV. LIST OF TOP PERFORMING ROIs AND THEIR CORRESPONDING NETWORKS

Classifier	Regions	Coordinates	Network	Accuracy (%)
CatBoost	45	-57.7, -40.6, 35.8	COTC	81.61
	12	-29, -35.9, -8.3	None	80.19
	143	39.2, -34.6, 57.5	SMH	77.71

	34	-23.4, -13.8, 64.2	SMH	77.61
	170	34, 24.4, 10	COTC	77.52
DT	88	-33.8, -33.2, -5.4	RST	83.04
	17	-5.5, 29.3, 44	FPTC	80.28
	105	-38.7, 4.8, 48.4	VA	80.28
	150	57.1, -17, -2.6	VA	80.28
	184	49, -54.5, 8.8	Visual	80.28

COTC-Cingulo-Opercular Task Control, SMH- Somatosensory Motor Hand, RST-Retro-Splenial Temporal network, FPTC-Fronto-parietal task control network, VA-Ventral Attention

IV. DISCUSSION

This research paper explores the development of a classification model for ASD. To address this objective, we put forth an approach that employs the CWT scalogram TF

technique, along with features such as GLCM, GLRLM, ZM, HM, FDTA, and FOS. These features are combined with ML classifiers, specifically CatBoost and DT. Additionally, a 5-fold cross-validation approach was employed to evaluate the system's performance. The results of the study hold promise for improving the accuracy and effectiveness of ASD classification models, enhancing their practical applicability. The results of the study could potentially set the stage for upcoming research aimed at creating more sophisticated and efficient classification systems for ASD. Rather than concentrating on individual subjects, our study takes a regional approach by examining various brain areas. We compared our study with the existing approach which is listed below in Table V.

TABLE V. LIST OF PREVIOUS STUDIES THAT USED TF METHODS ON FMRI BOLD SIGNALS FOR DIAGNOSTIC CLASSIFICATION OF ASD

Study	Database	Sites (subjects)	Atlas (Regions)	TF method	Feature extraction	Cross-validation	Classifier	Performance (%)	Network
Our study	ABIDE-I and ABIDE-II	OHSU (36 ASD, 35 TD)	Gordon's, HO, Diedrichsen (236)	CWT	GLCM, GLRLM, ZM, HM, FDTA and FOS	5-fold	CatBoost, DT	Accuracy - 83.04, Sensitivity - 66.66, Specificity- 85.57, F1 score -75.85	RST
[16]	ABIDE	- (41 ASD, 41 TD)	AAL (116)	PFT	Googlenet, DenseNet201, Resnet18, and Resnet101	5-fold, 10-fold, 15-fold and 20-fold	SVM, KNN	Accuracy- 96.7, Sensitivity- 96.6, Specificity- 96.9, Precision- 96.8	-
[17]	ABIDE	SDSU, STANFORD, UM, KKI, LEUVEN, SBL, TRINITY, CALTECH, CMU, MAX, NYU, OHSU, OLIN; PITT, UCLA, USM, YALE, (505 ASD, 530 TD)	AAL	WC, PWC FC	CNN	10-fold	CNN	Accuracy- 95.2, Sensitivity- 96.7, Specificity- 94.3, Precision- 94.8	-
[18]	ABIDE	CALTEC, CMU, KKI (41 ASD, 41 TD)	AAL (116)	CWT	Googlenet, DenseNet201, Resnet18, and Resnet101	No cross-validation (Training- 70%, Validation- 15% and testing- 15%)	SVM, KNN	Accuracy- 85.9, Sensitivity- 79.3, Specificity- 92.6	-
[19]	ABIDE	NYU, SBL, SDSU, TRINITY, YALE, USM, KKI, UM (36 ASD, 36 TD)	AAL (116)	WC	CNN	5-fold, 10-fold, 15-fold, 20-fold, leave-one site validation	CNN	Accuracy- 89.8, Sensitivity- 90.1, Specificity - 89.7, Precision- 89.6, F1 score-9.8	-
[6]	NDAR	- (50 ASD, 50 TD)	HO	CWT	CNN	4-fold	CNN	Accuracy-86, Sensitivity- 82, Specificity 92	CG, STG, PAC, AG

[20]	ABIDE	YALE, PITT, UCLA, SDSU, NYU, TRINITY (222 ASD, 246 TD)	HO (111)	D3TDWT	GARCH	5-fold	SVM	Accuracy-75.3	MTG, SG, PGR
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SVM-Support vector machine, KNN- K-nearest neighbours, PFT-Progressive Fourier Transform, CALTECH-California Institute of Technology, CMU- Carnegie Mellon University, NYU-New York University, SBL-Social Brain lab, SDSU- San Diego State University, Trinity-Trinity College Institute of Neuroscience, YALE-Yale School of Medicine, USM-University of Utah School of Medicine, KKI-Kennedy Krieger Institute, UM-University of Michigan, D3TDWT-Double-Density Dual-Tree Discrete Wavelet Transform, GARCH-Generalized autoregressive conditional heteroscedasticity, AAL-Automated anatomical labeling, HO- Harvard-Oxford, PITT- University of Pittsburgh School of Medicine, UCLA- University of California Los Angeles, STANFORD- Stanford University, UM-University of Michigan, LEUVEN-University of Leuven, MAX-Ludwig Maximilian University of Munich, OLIN-Olin Center, Institute of Living at Hartford Hospital, WC-Wavelet coherence, PWC-Principal wavelet coherence, MTG-Middle Temporal Gyri, SG-Supramarginal Gyrus, PGR-Paracingulate Gyrus R, CG-Cingulate gyri, STG-Superior temporal gyrus, PAC-Primary auditory cortex and AG-Angular gyrus

In this research, we accomplished a classification accuracy of 83.04%, a sensitivity of 66.66%, a specificity of 85.57%, and an F1 score of 75.85% by employing the CWT and DT classifier. Prior studies have achieved high classification accuracy by utilizing intricate CNN algorithms, either for feature extraction or classification, as evidenced by works such as [16][17][18][19] and [6]. However, these studies often fail to disclose the specific brain regions or networks contributing to diagnostic classification, with the exception of the study by [6]. Furthermore, our research makes use of resting state fMRI data sourced from the ABIDE database. This choice might not provide a fair ground for direct comparison with the investigation conducted by [6], which employs task-based fMRI data. Additionally, absence of robust cross-validation methods to authenticate their findings, akin to the observations made in the study conducted by [18]. Conversely, a study employing ML algorithms has reported lower classification accuracy when compared to our achieved results, as demonstrated by [20]. In conclusion, our study showcases a comparable performance while utilizing a restricted dataset and a focused set of features centered on the CWT technique, in conjunction with a computationally efficient ML algorithm.

V. LIMITATIONS AND FUTURE DIRECTIONS

The suggested approach for classifying ASD using time-series data through a combination of CWT-based TF analysis, texture, and shape features, along with ML classifiers, has demonstrated promising findings. While our results align with prior research, there are certain limitations to our proposed pipeline. Firstly, our study utilized fMRI data solely from a single site within the ABIDE database. Additionally, we exclusively employed the CWT-based TF method for analyzing fMRI BOLD signals. Lastly, our focus was on employing tree-based algorithms like CatBoost and DT to construct the diagnostic classification model. Despite these constraints, our research holds the potential for further exploration in future studies that could yield benchmark achievements in ASD diagnosis. Our investigation could be extended by encompassing alternative TF methods, more advanced feature extraction techniques, and incorporating deep learning algorithms.

VI. CONCLUSION

The research introduced an innovative method for diagnosing ASD using fMRI BOLD signal data. This involved the creation of scalogram images through the application of CWT. Notably, the incorporation of texture and shape-related features played a substantial role in effectively categorizing ASD cases. Interestingly, DT outperformed the CatBoost ML classifier in terms of achieving the highest accuracy. Our findings emphasize the

potential of utilizing texture and shape-based feature extraction methods in conjunction with ML classifiers to accurately diagnose ASD. However, further investigations are necessary to ascertain the adaptability and robustness of this approach across various demographics and scenarios. This study paves the way for future research endeavors in the realm of ASD diagnostic classification using fMRI BOLD signals, offering promising avenues for exploration.

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